

Fundamentals of Information Systems Security

Lab 1: Installing Kali Linux on VirtualBox

Objective: Set up a virtualized penetration testing environment.

Prerequisites

- A computer with at least 8 GB of RAM and 50 GB of free disk space.
- Oracle VirtualBox installed (<https://www.virtualbox.org>).
- Kali Linux ISO image downloaded from <https://www.kali.org/get-kali/>.

Learning Objectives

By the end of this lab, you will be able to:

- Create and configure a virtual machine in VirtualBox.
- Install Kali Linux from an ISO image.
- Perform basic system configuration and updates.
- Verify the installation and network connectivity.

Lab Steps

Step 1: Create a New Virtual Machine

1. Open **Oracle VM VirtualBox** and click **New**.
2. Set the following parameters:
 - **Name:** Kali Linux
 - **Folder:** Choose your preferred location.
 - **ISO Image:** Click the dropdown and select "Other" to browse and select your downloaded Kali Linux ISO.
 - **Type:** Linux
 - **Version:** Debian (64-bit)
3. Click **Next**.

Step 2: Allocate Resources

4. Allocate at least **2048 MB** of base memory (RAM). *Recommendation: 4096 MB or more for better performance.*
5. Allocate at least **20 GB** of virtual hard disk space. *Recommendation: 40 GB.*
6. Select **Create a virtual hard disk now** and click **Create**.
7. Choose **VDI (VirtualBox Disk Image)** and click **Next**.
8. Choose **Dynamically allocated** and click **Next**.
9. Set the disk size to **20 GB** (or more) and click **Create**.

Step 3: Configure System Settings (Optional Optimization)

10. Select the newly created VM and click **Settings**.
11. Go to **System** → **Processor** and allocate **2 or more CPUs** (if available).
12. Go to **Display** and increase **Video Memory** to at least **128 MB**.
13. Go to **Network** and ensure **Adapter 1** is set to **NAT** or **Bridged Adapter** (recommended for lab connectivity).

Step 4: Boot the Kali Linux Installer

14. Select the VM and click **Start**.
15. The VM will boot from the ISO. Select **Graphical Install**.
16. Follow the on-screen prompts:
 - **Language:** English (or your preference).
 - **Location:** Select your country.
 - **Keyboard:** Select your layout.

Step 5: Partitioning and Installation

17. When prompted for network configuration, enter a **Hostname** (e.g., `kali-lab`) and domain (leave blank).
18. Set a **root password** and create a **user account**.
19. For partitioning, select **Guided - use entire disk** and choose the virtual disk.
20. Select **All files in one partition** (recommended for beginners).
21. Confirm the changes and write the partition table to disk.
22. Wait for the base system installation to complete.

Step 6: Software Selection and Finalization

23. When the software selection screen appears, select the **Kali desktop environment** (Xfce is lightweight, GNOME is full-featured).
24. Ensure **Standard system utilities** is checked. Click **Continue**.
25. Wait for the software to install (this may take 15–30 minutes).
26. When prompted, choose **Yes** to install the GRUB boot loader to the master boot record.

Step 7: Reboot and First Login

27. Remove the installation media (or uncheck the ISO mount in VirtualBox) and reboot the VM.
28. Log in using the credentials you set during installation.
29. Once the desktop appears, you have successfully installed Kali Linux.

Post-Installation: System Update

Open a terminal and run the following commands to ensure your system is up to date:

```
sudo apt update
sudo apt full-upgrade -y
```

Optional: Install Guest Additions

To improve resolution and performance:

```
sudo apt install -y virtualbox-guest-x11
sudo reboot
```

Deliverables

Please submit a report containing:

1. A screenshot of your Kali Linux desktop showing the system info (`uname -a`).
2. A screenshot of the terminal output after running `ifconfig` or `ip a`.
3. A brief reflection on any challenges encountered during the installation process.